

A more modular approach is to split the code into a `Ball` class and a main program that instantiates a `Ball` object and makes calls to its methods. In this section we'll make this split, and I'll show you how to create multiple balls from the `Ball` class.

Creating a Ball Class

We'll start by extracting all code relating to the ball from the main program and moving it into a separate `Ball` class. Looking at the original code, we can see that the sections that deal with the ball are:

- Section #4, which loads the image of the ball
- Section #5, which creates and initializes all the variables that have something to do with the ball
- Section #8, which includes code for moving the ball, detecting an edge bounce, and changing speed and direction
- Section #10, which draws the ball

From this we can conclude that our `Ball` class will require the following methods:

`create()` Loads an image, sets a location, and initializes all instance variables

`update()` Changes the location of the ball in every frame, based on the x speed and y speed of the ball

`draw()` Draws the ball in the window

The first step is to create a project folder, in which you need a *Ball.py* for the new `Ball` class, the main code file *Main_BallBounce.py*, and an *images* folder containing the *ball.png* image file.

[Listing 6-1](#) shows the code of the new `Ball` class.